



REVIEW

Cardiovascular Disease and Psoriasis

Małgorzata Ponikowska^{ID} · Loreena Hill^{ID} · Christopher S. Lee^{ID} · Michela Barisone^{ID} ·
Piotr Ponikowski^{ID} · Heba M. Aldossary^{ID} · Maria Jędrzejczyk^{ID} · Grzegorz Kubielas^{ID} ·
Izabella Uchmanowicz^{ID}

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ABSTRACT

Psoriasis is a chronic inflammatory skin disease characterized by well-demarcated erythematous plaques with silvery scales that affects 2–3% of the global population. Beyond its dermatological manifestations, psoriasis has recently been

recognised as a significant cardiovascular risk factor, patients with psoriasis have an approximately 50% increased relative risk of major cardiovascular events compared with the general population. This review examines the complex relationship between psoriasis and cardiovascular disease, exploring the epidemiological evidence, underlying pathophysiological mechanisms, clinical implications and therapeutic considerations. The

M. Ponikowska
University Centre of General Dermatology
and Oncodermatology, Faculty of Medicine,
Wrocław Medical University, Wrocław, Poland

L. Hill
School of Nursing and Paramedic Science, Ulster
University, Londonderry, UK

C. S. Lee
Boston College William F. Connell School
of Nursing, Chestnut Hill, MA, USA

C. S. Lee
Department of Emergency Medical Service, Faculty
of Nursing and Midwifery, Wrocław Medical
University, Wrocław, Poland

M. Barisone
Clinical Epidemiology and Research Center, IRCCS
Humanitas Research Hospital, Pieve Emanuele, MI,
Italy

P. Ponikowski
Institute of Heart Diseases, University Hospital
of Wrocław Medical University, Wrocław, Poland

H. M. Aldossary (✉)
Department of Nursing, Prince Sultan Military

College of Health Sciences, Dhahran, Saudi Arabia
e-mail: heba@psmchs.edu.sa

H. M. Aldossary
Frances Payne Bolton School of Nursing at Case
Western Reserve University, Cleveland, OH, USA

M. Jędrzejczyk · I. Uchmanowicz
Division of Research Methodology, Department
of Nursing, Faculty of Nursing and Midwifery,
Wrocław Medical University, Wrocław, Poland

G. Kubielas
Department of Health Care Organization,
Department of Nursing, Faculty of Nursing
and Midwifery, Wrocław Medical University,
Wrocław, Poland

G. Kubielas
Department of Health Care Services, Polish National
Health Fund, Central Office in Warsaw, Warsaw,
Poland

I. Uchmanowicz
Centre for Cardiovascular Health, Edinburgh Napier
University, Sighthill Campus, Edinburgh, UK

inflammatory milieu characteristic of psoriasis, involving T cell activation, cytokine dysregulation and systemic inflammation, creates a pro-atherogenic environment that accelerates cardiovascular disease development. Understanding the mechanisms of cardiovascular risk is crucial for clinicians managing psoriatic patients, as it necessitates comprehensive risk assessment and preventive strategies beyond traditional dermatological care.

Keywords: Psoriasis; Cardiovascular disease; Inflammation; Atherosclerosis; Cytokines; Risk factors

Key Summary Points

Psoriasis is now widely recognized as a systemic inflammatory disease that significantly increases the risk of cardiovascular morbidity and mortality.

Shared inflammatory mechanisms between psoriasis and atherosclerosis, including cytokine dysregulation and endothelial dysfunction, contribute to a pro-atherogenic environment.

Current cardiovascular risk assessment tools often underestimate risk in patients with psoriasis, highlighting the need for psoriasis-specific modifications and biomarkers.

Anti-inflammatory treatments, particularly biologic therapies, show promise in reducing cardiovascular risk, but data from randomized controlled trials remain inconclusive.

Integrated care models involving dermatology, cardiology and primary care are essential to improve cardiovascular outcomes in patients with psoriasis.

INTRODUCTION

Psoriasis is one of the most prevalent chronic immune-mediated inflammatory skin diseases, affecting approximately 2–3% of the global

population, or about 125 million individuals worldwide [1]. Clinically, it is characterised by well-demarcated erythematous plaques covered with silvery scales, resulting from keratinocyte hyperproliferation and immune cell infiltration. Disease severity is routinely assessed using validated tools such as the Psoriasis Area and Severity Index (PASI), Body Surface Area (BSA) and Dermatology Life Quality Index (DLQI), which together guide therapeutic decisions and stratify patients into mild, moderate or severe categories [2]. Beyond skin involvement, psoriasis is a heterogeneous disease, with up to 25% of patients developing psoriatic arthritis (PsA), a chronic inflammatory arthropathy manifesting as peripheral arthritis, enthesitis, dactylitis or spondylitis. PsA contributes an additional systemic inflammatory burden, further increasing cardiovascular risk in affected individual [3].

Contemporary systemic and biologic therapies have dramatically improved outcomes in psoriasis, but their adverse effect profiles warrant explicit consideration in any discussion of comorbidity management. Immunosuppression associated with many systemic agents increases susceptibility to infections and may influence malignancy risk. Conventional systemic drugs such as cyclosporine are associated with hypertension, nephrotoxicity and dyslipidaemia, which can exacerbate cardiovascular risk, while methotrexate carries well-recognised hepatotoxicity and haematologic adverse events. Some biologic agents – although generally well tolerated – have specific safety signals to monitor, for example, tumour necrosis factor (TNF) inhibitors have been infrequently associated with worsening or de novo heart failure in selected patients. These safety considerations are relevant both for treatment selection and for the intensity of cardiovascular monitoring, emphasising the need for multidisciplinary decision-making when systemic therapy is contemplated [4–6].

Over the past decade, the understanding of psoriasis has evolved considerably, and it is now widely recognised as a systemic inflammatory disorder with far-reaching implications. Large epidemiological studies, including those by Gelfand et al. and Mehta et al., have shown that patients with psoriasis are at significantly increased risk of major cardiovascular events, particularly

myocardial infarction and coronary artery disease. These findings have been incorporated into contemporary management recommendations such as the Living EuroGuiDerm Guideline, which emphasises the importance of early cardiovascular risk assessment in psoriatic patients [2].

Further evidence of this association comes from a prospective study by Piaserico et al. [7], which demonstrated that nearly one-third of 503 patients with severe psoriasis had coronary microvascular dysfunction despite being asymptomatic. This has fundamentally reshaped clinical practice, underscoring the need for a multidisciplinary approach involving dermatologists, cardiologists and primary care physicians.

The 2019 ACC/AHA Guideline on the Primary Prevention of Cardiovascular Disease now lists psoriasis as a cardiovascular ‘risk-enhancing’ factor [8], reflecting a paradigm shift from viewing psoriasis purely as a dermatological condition to recognising its systemic inflammatory and pro-atherogenic nature. This association carries major clinical implications, as cardiovascular disease remains the leading cause of mortality worldwide [9]. For patients with moderate to severe psoriasis, the elevated cardiovascular risk translates into earlier onset of coronary events and measurable reductions in both life expectancy and quality of life [10].

Aim

The aim of this comprehensive review is to synthesise current evidence on the relationship between psoriasis and cardiovascular disease, with specific objectives to: examine the epidemiological evidence establishing psoriasis as a cardiovascular risk factor; elucidate the pathophysiological mechanisms linking chronic inflammatory skin disease to accelerated atherosclerosis; analyse contemporary approaches to cardiovascular risk assessment, stratification and therapeutic strategies for risk reduction in psoriatic populations, including both traditional risk factor management and anti-inflammatory approaches; review modern guidelines and clinical recommendations for integrated care; and identify future research priorities and emerging therapeutic targets.

This review not only synthesises current evidence but also serves as a practical tool for clinicians, offering evidence-based insights to support cardiovascular risk assessment, integrated management and informed decision-making in patients with psoriasis.

Ethics Statement

This article is based on previously conducted studies and does not contain any new studies with human participants or animals performed by the authors. Therefore, ethical approval and informed consent were not required.

EPIDEMIOLOGY AND CLINICAL EVIDENCE

Contemporary Risk Quantification

Recent analyses confirm that psoriasis is associated with a clinically significant increase in cardiovascular risk. In the meta-analysis by Miller et al. [11], which included over 300,000 patients with up to 10 years of follow-up, psoriasis was linked to higher odds of cardiovascular disease, notably ischemic heart disease and myocardial infarction. Samarasekera et al. [12] similarly reported hazard ratios ranging from approximately 1.1 to 3.0, depending on disease severity and outcomes such as stroke and myocardial infarction, affirming psoriasis as an independent cardiovascular risk factor over a mid- to long-term timeline.

Importantly, newer meta-analytic data refine these findings. A 2022 meta-analysis of 31 cohort studies – including over 665,000 patients with psoriasis and nearly 18 million controls with follow-up up to around 10 years – found pooled relative risks of approximately 1.17 for myocardial infarction, 1.19 for stroke, 1.46 for cardiovascular mortality and higher estimates in individuals with moderate-to-severe disease, reinforcing the sustained long-term cardiovascular impact of psoriasis [13].

Cardiovascular Event Spectrum

The increased cardiovascular risk in psoriasis encompasses a wide range of clinical outcomes. Severe psoriasis has been linked to up to a three-fold increase in myocardial infarction, approximately 60% higher odds of stroke and about 40% higher odds of cardiovascular mortality, with risk magnified by greater disease severity [12]. This pattern reflects the systemic inflammatory burden of psoriasis, a chronic immune-mediated disease that can be classified by severity using metrics such as BSA, PASI and DLQI [1, 2].

In addition to major cardiovascular events, heart failure incidence rises with psoriasis severity, as demonstrated by Khalid et al. [14] in a nationwide Danish registry study. Subclinical atherosclerosis is also more prevalent: Mansouri et al. [15] reported that patients with psoriasis have approximately double the odds of coronary artery calcification compared with matched controls, with atherosclerotic burden approaching that seen in patients with type 2 diabetes, not a diabetic-only control group.

Furthermore, advanced coronary CT angiography by Lerman et al. [16] revealed a 15% greater non-calcified coronary plaque burden and more high-risk plaque features in psoriatic patients, correlating directly with skin disease severity.

Psoriatic Arthritis and Enhanced Risk

Approximately 25% of patients with psoriasis develop PsA, a chronic inflammatory musculoskeletal disease manifesting as peripheral arthritis, enthesitis, dactylitis and spondylitis. PsA constitutes a subgroup with particularly elevated cardiovascular risk. Eder et al. [3] demonstrated that patients with PsA have a significantly higher carotid plaque burden compared with those with skin-limited psoriasis. Furthermore, contemporary registry analyses confirm that PsA confers cardiovascular risk comparable to severe cutaneous psoriasis, suggesting additive effects of both joint and skin inflammation on cardiovascular disease development [17].

PATHOPHYSIOLOGICAL MECHANISMS

Inflammatory Cascade and Molecular Pathways

Recent research has elucidated key molecular mechanisms linking psoriasis to increased cardiovascular risk. Mehta et al. [18] performed transcriptomic analyses comparing psoriatic skin lesions with atherosclerotic plaques, uncovering significant molecular overlap, particularly in interferon-gamma (IFN- γ) and tumour necrosis factor-alpha (TNF- α) mediated pathways. These inflammatory pathways synergistically activate vascular endothelial cells, inducing substantial increases – over 5000-fold – in pro-atherogenic transcripts such as vascular cell adhesion molecule-1 (VCAM-1) and chemokine (C-X-C motif) ligand 10 (CXCL10).

More recent studies have identified inflammasome signaling as a central mechanistic link. Garshick et al. (2019) demonstrated that inflammasome activation represents the most significantly differentially expressed systemic pathway in psoriasis and plays a causal role in atherosclerosis development [19]. This insight highlights potential therapeutic targets that may simultaneously reduce psoriatic disease activity and cardiovascular risk.

Endothelial Dysfunction and Vascular Biology

Recent studies have directly investigated vascular endothelial function in patients with psoriasis. Garshick et al. (2020) analysed endothelial cells isolated from patients with psoriasis and found a two- to eightfold upregulation of pro-inflammatory transcripts compared with patients without psoriasis. These transcripts include VCAM-1, interleukin-1 beta (IL-1 β) and cyclooxygenase-2 (COX-2) [20]. This expression profile closely resembles that of endothelial cells stimulated by combined action of TNF- α , interleukin-17A (IL-17A) and IFN- γ , demonstrating a direct vascular impact of psoriatic inflammation.

Advanced assessments of vascular function using flow-mediated dilation and arterial stiffness measurements have consistently shown impaired endothelial function in patients with psoriasis compared with healthy controls. Moreover, the degree of endothelial dysfunction correlates with both psoriasis severity and systemic inflammatory markers [21].

Cellular and Humoral Immune Mechanisms

Several specific cellular immune components contributing to psoriasis-associated cardiovascular risk have been identified. Teague et al. [22] characterised neutrophil subsets in psoriasis, identifying that low-density granulocytes are 30% more prevalent in patients with psoriasis and are associated significantly with coronary atherosclerosis burden. These specialised neutrophils induce enhanced endothelial damage through neutrophil extracellular trap formation. Furthermore, studies have revealed significant activation of platelets in patients with psoriasis. Garshick et al. [23] demonstrated that psoriatic platelets exhibit interferon signatures and elevated cyclooxygenase-1 (COX-1) expression correlating with skin severity. Platelets from patients with psoriasis induce >20-fold increases in endothelial pro-atherosclerotic transcripts, suggesting a potential therapeutic targets for antiplatelet therapy.

Emerging Molecular Pathways

In addition to the well-characterized TNF- α , IL-17 and IFN- γ pathways, recent studies highlight novel inflammatory mediators linking psoriasis to cardiovascular disease. The IL-36 cytokine family, part of the IL-1 superfamily, plays a pivotal role in amplifying keratinocyte activation and innate immune responses; elevated IL-36 activity has been implicated in systemic inflammation and may contribute to vascular injury and atherogenesis [24].

Members of the S100 protein family – particularly S100A7A and the heterodimer S100A8/A9 – act as alarmins driving neutrophil and monocyte recruitment; these proteins correlate with

psoriasis severity and have been associated with plaque instability in atherosclerosis [25, 26].

Another emerging factor is GPR15, a G-protein coupled receptor involved in epithelial immunity and lymphocyte trafficking; altered GPR15 expression has been observed in inflammatory conditions and may link mucosal/epithelial immune activation to systemic vascular inflammation [27].

Transcriptomic Overlap Between Psoriatic Lesions and Atherosclerotic Plaques

Transcriptomic studies of psoriatic skin have revealed a substantial molecular overlap with atherosclerotic lesions, highlighting shared upregulation of interferon signaling, TNF-related pathways and IL-17/IL-23 axis components. These transcriptomic signatures include genes involved in leukocyte recruitment, endothelial activation and lipid metabolism, providing mechanistic evidence that chronic systemic inflammation in psoriasis directly contributes to vascular pathology. Integrating transcriptomic data therefore strengthens the hypothesis that dyslipidaemia and metabolic syndrome observed in psoriasis are, at least in part, a consequence of persistent inflammatory signaling rather than merely co-existent conditions [26, 28–30].

TRADITIONAL CARDIOVASCULAR RISK FACTORS IN CONTEMPORARY PSORIATIC POPULATIONS

Prevalence and Modern Recognition Patterns

Contemporary studies reveal concerning patterns of cardiovascular risk factor prevalence and management in psoriatic populations. Kimball et al. [31] analysed data from approximately 3000 patients with psoriasis enrolled in modern clinical trials. The average age of participants was 45 years, and the study found that 59% of patients had at least two traditional cardiovascular risk factors, while 29% had three or more [31]. These statistics are particularly concerning

given the relatively young average age of the study population.

Recent analyses also demonstrate persistent under-recognition and under-treatment of cardiovascular risk factors among patients with psoriasis. The PSO-RISK study by Cea-Calvo et al. [32] revealed that approximately 20% of patients with psoriasis had diagnosed diabetes and under-treated hypertension, while nearly 40% of patients with hyperlipidaemia lacked appropriate therapy. Among those receiving treatment, about 60% were not prescribed evidence-based medications at guideline-recommended targets, indicating a significant care gap.

Contemporary Dyslipidaemia Patterns

Modern lipid analyses have revealed characteristic dyslipidaemic patterns in patients with psoriasis. Mehta et al. [33] utilised nuclear magnetic resonance spectroscopy to demonstrate increased LDL particle concentration and decreased LDL particle size in patients with psoriasis. Holzer et al. [34] demonstrated that psoriatic patients have up to 15% lower HDL cholesterol levels and nearly 80% reduced HDL efflux capacity. Additional studies have identified oxidative lipid modifications as key mechanistic factors. Sorokin et al. [35] demonstrated that patients with psoriasis have 15% higher oxidized HDL levels, with oxidised lipoprotein levels correlating directly with non-calcified coronary plaque burden. These findings provide mechanistic insights into accelerated atherosclerosis in psoriatic populations.

Contemporary evidence supports aggressive cardiovascular risk factor management in psoriatic populations. Wu et al. [36] conducted a large retrospective analysis of approximately 9000 patients with psoriasis and found that statin therapy was associated with significant reductions in incident myocardial infarction (hazard ratio 0.31 [0.22–0.43]). Recent post hoc analyses of major lipid trials have confirmed efficacy in psoriatic populations. Ports et al. [37] analysed secondary prevention trials including psoriatic patients, confirming that high-intensity statins achieve similar lipid reductions and

cardiovascular event reductions compared with non-psoriatic patients. Contemporary research has also identified elevated PCSK9 in psoriatic patients, suggesting potential benefits of PCSK9 inhibition [38].

Hypertension and Metabolic Dysfunction

Studies have clarified mechanistic links between psoriatic inflammation and the development of hypertension. Takeshita et al. [39] demonstrated, using large-scale population data, that the severity of hypertension correlates positively with psoriasis disease severity. Experimental work by Karbach et al. [40] further revealed that overexpression of IL-17A directly induces hypertension via mechanisms involving vascular dysfunction.

Additionally, the prevalence of metabolic syndrome in patients with psoriasis exhibits a clear dose–response relationship with disease severity. In a large UK population study, Langan et al. [41] found that patients with mild psoriasis had a 22% increased odds of metabolic syndrome, while this risk increased to 56% in moderate and 98% in severe psoriasis.

CLINICAL ASSESSMENT AND CONTEMPORARY RISK STRATIFICATION

Modern Cardiovascular Risk Calculation

Contemporary guidelines have formally recognised the limitations of traditional cardiovascular risk calculators in patients with psoriasis. The 2019 American College of Cardiology/American Heart Association (ACC/AHA) Primary Prevention Guidelines specifically identify psoriasis as a cardiovascular risk enhancer, recommending that it be taken into account when assessing patients' 10-year cardiovascular risk [8]. This represents a Class IIa recommendation with Level of Evidence B-NR (non-randomised data).

Similarly, the 2019 joint guidelines from the American Academy of Dermatology (AAD) and the National Psoriasis Foundation (NPF) provide

specific implementation strategies. They recommend applying a 1.5 multiplication factor to the calculated 10-year cardiovascular risk scores in patients with more than 10% BSA affected by psoriasis or those requiring systemic therapy [42]. This evidence-based approach aims to correct the systematic underestimation of cardiovascular risk in populations with inflammatory diseases.

Contemporary Biomarkers and Advanced Assessment

Biomarker research has identified novel cardiovascular risk indicators specific to psoriatic populations. Joshi et al. [43] identified glycoprotein acetylation (GlycA) as a novel inflammatory biomarker that improves coronary atherosclerosis prediction in psoriasis beyond traditional risk factors and hs-CRP. This represents a significant advancement in psoriasis-specific risk stratification. Imaging approaches also have added considerably to subclinical disease detection. Naik et al. [44] demonstrated correlations between psoriasis skin severity and pan-arterial vascular inflammation using [18F]-fluorodeoxyglucose positron emission tomography. However, it is important to note that despite their diagnostic value, these advanced biomarkers and imaging approaches remain prohibitively expensive and are not widely accessible in routine clinical practice, particularly in healthcare systems.

Beyond traditional markers such as hs-CRP, novel psoriasis-specific biomarkers are emerging with potential relevance for cardiovascular disease. S100 proteins, particularly S100A8/A9 and S100A12, reflect neutrophil activation and correlate with both psoriasis severity and vascular inflammation. Chemokines, such as CXCL8 and CXCL10, as well as elevated IL-17A and IL-22 levels, have also been implicated as potential predictors of subclinical atherosclerosis. Incorporating such markers into clinical practice could improve risk stratification, although their utility requires validation in large prospective cohorts [45].

Modern Screening Recommendations

Current guidelines from both dermatology and cardiology societies emphasise the need for comprehensive cardiovascular risk screening in patients with psoriasis. The 2019 American Academy of Dermatology–National Psoriasis Foundation (AAD–NPF) guidelines recommend regular evaluation of blood pressure, lipid profile, glucose/HbA1c, body mass index and smoking status, with screening frequency guided by disease severity, duration and systemic treatment use [42]. The 2019 American College of Cardiology/American Heart Association (ACC/AHA) prevention guidelines formally recognise psoriasis as a cardiovascular risk-enhancing factor, supporting more vigilant risk estimation and early intervention [8].

Dermatology guidelines place particular emphasis on dermatologist-initiated screening and coordination with primary care or cardiology, while cardiology guidelines highlight psoriasis mainly as an additional atherosclerotic risk enhancer [2, 46]. Comprehensive screening thus involves both traditional risk factor assessment and awareness of psoriasis-specific modifiers, ensuring timely prevention strategies.

CONTEMPORARY THERAPEUTIC INTERVENTIONS AND MANAGEMENT STRATEGIES

Evidence-Based Lifestyle Modifications

Systematic reviews confirm the effectiveness of lifestyle interventions in psoriasis management. A meta-analysis by Upala and Sanguankeo [47] including ~900 overweight or obese patients demonstrated that weight reduction through caloric restriction improved psoriasis severity, with up to a threefold increase in clearance rates when combined with standard therapy. Contemporary dietary research emphasizes anti-inflammatory patterns. A systematic review for the National Psoriasis Foundation by Ford et al. [48] found moderate evidence supporting Mediterranean-style

diets, correlating with reduced skin severity and lower systemic inflammation.

Contemporary Anti-Inflammatory Therapy for Cardiovascular Risk

Modern research has extensively investigated whether treating psoriatic inflammation can mitigate cardiovascular risk. Observational studies consistently suggest benefit, with Wu et al. [36] reporting approximately 50% reductions in myocardial infarction incidence among patients receiving biologic therapies, such as TNF- α inhibitors, IL-17 inhibitors and interleukin-23 (IL-23) inhibitors, compared with those treated with topical regimens.

Surrogate cardiovascular endpoints have also been explored. Dey et al. [49] demonstrated that biologic treatment was associated with an $\approx$ 8% reduction in aortic vascular inflammation and a $\approx$ 6–7% decrease in coronary plaque burden over 12 months in patients with moderate-to-severe psoriasis. Elnabawi et al. [50] confirmed these findings in larger prospective cohorts.

However, recent randomized placebo-controlled trials have yielded mixed results. Mehta et al. [51] found that treatment with the biologic adalimumab or narrow-band UVB phototherapy did not significantly reduce vascular inflammation despite adequate skin clearance. Similarly, Gelfand et al. [52] reported no meaningful change in aortic inflammation with secukinumab treatment. These findings highlight the complexity of translating reductions in psoriatic inflammation into consistent cardiovascular benefits.

Guidelines and Clinical Recommendations

Multiple professional societies have updated their guidelines to incorporate recent evidence regarding cardiovascular risk in patients with psoriasis. The 2019 American College of Cardiology/American Heart Association (ACC/AHA) Primary Prevention Guidelines mark a significant milestone by explicitly recognizing psoriasis as a cardiovascular risk enhancer and recommending that clinicians consider it when assessing a

patient's 10-year cardiovascular risk [8]. This recommendation is categorized as Class IIa with Level of Evidence B-NR, underscoring its clinical importance.

In dermatology, the 2019 Joint American Academy of Dermatology (AAD) and National Psoriasis Foundation (NPF) guidelines provide comprehensive, evidence-based recommendations aimed at dermatologists managing patients with psoriasis [42]. These guidelines emphasize systematic cardiovascular risk factor screening, patient education about the increased cardiovascular risk associated with psoriatic disease and structured collaboration with cardiovascular specialists. The guidance also includes specific strategies for risk enhancement identification and treatment thresholds based on contemporary data.

Integrating cardiovascular risk assessment and management into psoriasis care is essential for optimizing patient outcomes. Whether these guidelines are presented as a separate section or integrated within therapeutic discussions depends on the scope and focus of the work; however, highlighting them ensures clinicians remain informed about the evolving standard of care.

IMPLEMENTATION OF CONTEMPORARY CARE MODELS

Despite the increased cardiovascular risk, many patients with psoriasis remain undertreated for risk factors such as diabetes, hypertension and hyperlipidemia. There is a growing demand for a multidisciplinary approach involving dermatologists in the assessment and management of cardiovascular risk [46]. Modern healthcare delivery increasingly emphasizes integrated care models that coordinate dermatologic and cardiovascular management through shared care pathways, electronic health records and standardized screening protocols.

Cardio-dermatology programs have emerged at several academic centres to address this need. For example, the multidisciplinary psoriasis clinic at the University of Pennsylvania integrates dermatologists, cardiologists and other

specialists to provide coordinated cardiovascular risk assessment and management alongside dermatologic care [53].

These programs facilitate the implementation of evidence-based guidelines and promote research into optimal management strategies, improving patient outcomes through enhanced collaboration.

CONTEMPORARY RESEARCH PRIORITIES AND FUTURE DIRECTIONS

Current Clinical Trial Landscape

Current priorities in the field focus on conducting large-scale, randomized controlled trials with clinical cardiovascular endpoints. Contemporary study designs emphasize adequate sample sizes and sufficient follow-up durations to capture meaningful cardiovascular events such as myocardial infarction and stroke. Several ongoing trials are investigating whether treating psoriatic inflammation reduces cardiovascular event risk, addressing important gaps in current knowledge [54].

Future research should also aim to determine optimal targets for cardiovascular risk factors specifically in psoriatic populations, as current recommendations largely extrapolate from general population guidelines without psoriasis-specific validation. Key research priorities include evaluating whether more aggressive management of lipids, blood pressure and glycemic control confers additional benefit in high-risk patients with psoriasis.

Modern Biomarker Development and Precision Medicine

Contemporary research increasingly focuses on advanced biomarkers beyond traditional cardiovascular risk factors. Glycoprotein acetylation, inflammatory cytokine panels and genetic risk scores represent promising areas for development [43]. The identification of biomarkers predicting both cardiovascular risk and treatment

response could offer a more personalised approach to care. Modern genomic and transcriptomic approaches also may offer opportunities for precision medicine implementation. Contemporary studies utilising multi-omics approaches may identify psoriatic patients at highest cardiovascular risk and most likely to benefit from specific interventions.

Therapeutic Innovation and Target Identification

Contemporary therapeutic development focuses on pathways shared between psoriatic inflammation and atherosclerosis. The inflammasome pathway, representing the highest differentially expressed systemic pathway in psoriasis, offers promising therapeutic targets for dual benefit [19]. Modern drug development increasingly emphasises precision approaches utilising genetic, transcriptomic and proteomic data to optimise therapeutic selection. Such approaches could maximise clinical benefit while minimising adverse effects and healthcare costs.

Long Non-Coding RNAs

Long non-coding RNAs (lncRNAs) are emerging regulators of immune, metabolic and vascular pathways. Recent genome-wide analyses identified differentially expressed lncRNAs in psoriasis that modulate inflammatory gene networks and may influence endothelial function and lipid metabolism; therefore, lncRNAs represent promising molecular biomarkers and potential therapeutic targets at the psoriasis–cardiovascular disease interface.

CONTEMPORARY MONITORING AND CLINICAL INTEGRATION

Modern practice for patients with psoriasis requires integrated cardiovascular risk monitoring that considers both traditional risk factor evolution and psoriatic disease activity.

Contemporary clinical practice emphasises systematic assessment during routine dermatological visits, utilising electronic health record prompts and standardised screening protocols. The integration of cardiovascular risk monitoring into routine dermatological care provides opportunities for early intervention and prevention. Contemporary approaches emphasise team-based care models with coordinated management between dermatologists, primary care providers and cardiovascular specialists.

HEALTHCARE SYSTEM AND ECONOMIC CONSIDERATIONS

Contemporary Healthcare Utilization Patterns

Recognition of elevated cardiovascular risk in patients with psoriasis has significant implications for modern healthcare resource utilization. Early identification and aggressive management of cardiovascular risk factors may prevent costly future cardiovascular events, potentially offsetting increased screening and treatment costs. Healthcare delivery requires multidisciplinary care coordination across specialties and geographic boundaries, which adds complexity but improves patient outcomes. Modern electronic health record systems and care coordination platforms can facilitate better communication between providers and ensure comprehensive risk management.

Modern Cost-Effectiveness Considerations

Patients with psoriasis incur substantial direct healthcare costs and indirect costs from productivity losses, which increase with disease severity and treatment intensity [55]. Contemporary economic analyses of cardiovascular risk management strategies in patients with psoriasis remain limited but do represent an important area of research. Cost-effectiveness of enhanced screening, aggressive risk factor modification and anti-inflammatory therapy for cardiovascular risk reduction requires evaluation to guide modern healthcare policy and resource allocation.

Economic evaluations of psoriasis treatment show systemic therapies (including biologics) have higher upfront costs but may be justified by improved quality of life and reduced disease burden. Similarly, cardiovascular prevention programs may increase short-term spending but yield long-term saving and health benefits [56].

PATIENT EDUCATION AND CONTEMPORARY SHARED DECISION MAKING

Modern Risk Communication Strategies

Effective patient education regarding cardiovascular risk requires clear, evidence-based communication that conveys both the significance of the risk and available interventions. This communication should be tailored to the unique needs of patients with psoriasis to enhance understanding and support informed decision-making.

Contemporary clinical practice increasingly emphasizes ongoing risk communication throughout the course of chronic disease. Digital health platforms and patient portals play a key role in this process by providing continuous access to educational materials and enabling patient engagement. Studies have shown that both patients and providers generally hold positive attitudes toward the use of such portals for chronic disease management, facilitating better communication and adherence [57].

Current Treatment Adherence and Engagement

Modern management of both psoriatic disease and cardiovascular risk factors presents adherence challenges requiring systematic approaches. Current healthcare delivery emphasises shared decision-making approaches involving patients, and if necessary caregivers, in treatment planning and goal-setting to improve adherence and outcomes. Decision aids have been shown to improve patient knowledge, risk perception and alignment of treatment with personal values,

thereby enhancing engagement and adherence to treatment plans [58]. Patient preference assessment, side effect tolerance evaluation and individual risk tolerance guidance should inform treatment recommendations within evidence-based frameworks using contemporary decision support tools.

Patient Education

A patient education program in psoriasis and cardiovascular disease management focused on increasing patient knowledge about the link between psoriasis and elevated cardiovascular risk, improving self-management and encouraging coordinated care to reduce cardiovascular disease complications. The results of a cross-sectional study show that on 102 patients interviewed with psoriasis underline the incomplete knowledge of the association between psoriasis and cardiovascular disease for which better patient education is needed. Greater awareness of cardiovascular risk may encourage patients to undergo screening and management of cardiovascular risk factors and to adopt healthy lifestyles [59]. The US National Psoriasis Foundation (NPF) supports research and initiatives to translate cardiovascular prevention guidelines into practice for patients with psoriasis. Their approach involves care coordination embedded in patient navigator services to address barriers and improve screening and management of cardiovascular risk factors in this population [60].

CONCLUSIONS AND CONTEMPORARY PERSPECTIVES

The relationship between psoriasis and cardiovascular disease represents a well-established paradigm of how chronic inflammatory conditions extend beyond primary organ involvement to affect overall health and mortality. Contemporary evidence strongly supports psoriasis as a legitimate cardiovascular risk enhancer requiring clinical attention and systematic management approaches.

Current evidence supports aggressive management of traditional cardiovascular risk

factors in patients who have psoriasis, with consideration of enhanced risk assessment and potentially more aggressive treatment targets based on recent clinical trial data. The role of anti-inflammatory therapy for cardiovascular risk reduction remains promising but requires definitive proof through future and ongoing large-scale randomised controlled trials.

Modern healthcare providers caring for patients with psoriasis must adopt comprehensive approaches addressing both dermatological and cardiovascular health. This requires contemporary multidisciplinary collaboration, evidence-based patient education and systematic approaches to cardiovascular risk assessment and management. This review is intended to support these efforts as a practical clinical resource, translating complex epidemiological and mechanistic evidence into actionable strategies for everyday practice.

The future of psoriatic patient care lies in precision medicine approaches considering individual risk profiles, treatment responses and patient preferences. Continued research into biomarkers, therapeutic targets and care delivery models will refine our ability to prevent cardiovascular events and improve overall health outcomes in this high-risk population using contemporary evidence-based strategies.

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Declarations

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REFERENCES

1. Parisi R, Symmons DPM, Griffiths CEM, Ashcroft DM. Global epidemiology of psoriasis: a systematic review of incidence and prevalence. *J Invest Dermatol*. 2013;133(2):377–85.
2. Nast A, Smith C, Spuls PI, Avila Valle G, Bata-Csörgő Z, Boonen H, et al. Euroguiderm guideline on the systemic treatment of Psoriasis vulgaris—part 1: treatment and monitoring recommendations. *J Eur Acad Dermatol Venereol*. 2020;34(11):2461–98.
3. Eder L, Jayakar J, Shanmugarajah S, Thavaneswaran A, Pereira D, Chandran V, et al. The burden of carotid artery plaques is higher in patients with psoriatic arthritis compared with those with psoriasis alone. *Ann Rheum Dis*. 2013;72(5):715–20.
4. Chung ES, Packer M, Lo KH, Fasanmade AA, Willerson JT. Randomized, double-blind, placebo-controlled, pilot trial of Infliximab, a chimeric monoclonal antibody to tumor necrosis factor- α , in patients with moderate-to-severe heart failure: results of the anti-TNF therapy against congestive heart failure (ATTACH) trial. *Circulation*. 2003;107(25):3133–40.
5. Hoorn EJ, Walsh SB, McCormick JA, Zietse R, Unwin RJ, Ellison DH. Pathogenesis of calcineurin inhibitor-induced hypertension. *J Nephrol*. 2012;25(3):269–75.
6. Montaudie H, Sbidian E, Paul C, Maza A, Gallini A, Aractingi S, et al. Methotrexate in psoriasis: a systematic review of treatment modalities, incidence, risk factors and monitoring of liver toxicity. *J Eur Acad Dermatol Venereol*. 2011;25(s2):12–8.
7. Piaserico S, Papadavid E, Cecere A, Orlando G, Theodoropoulos K, Katsimbri P, et al. Coronary microvascular dysfunction in asymptomatic patients with severe psoriasis. *J Invest Dermatol*. 2023;143(10):1929–1936.e2.
8. Arnett DK, Blumenthal RS, Albert MA, Buroker AB, Goldberger ZD, Hahn EJ, et al. 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Circulation* [Internet]. 2019. <https://doi.org/10.1161/CIR.0000000000000677>.
9. Benjamin EJ, Muntner P, Alonso A, Bittencourt MS, Callaway CW, Carson AP, et al. Heart disease and stroke statistics—2019 update: a report from the American Heart Association. *Circulation* [Internet]. 2019. <https://doi.org/10.1161/CIR.0000000000000659>.

10. Karbach S, Hobohm L, Wild J, Münzel T, Gori T, Wegner J, et al. Impact of psoriasis on mortality rate and outcome in myocardial infarction. *J Am Heart Assoc.* 2020;9(18):e016956.
11. Miller IM, Ellervik C, Yazdanyar S, Jemec GBE. Meta-analysis of psoriasis, cardiovascular disease, and associated risk factors. *J Am Acad Dermatol.* 2013;69(6):1014–24.
12. Samarasekera EJ, Neilson JM, Warren RB, Parnham J, Smith CH. Incidence of cardiovascular disease in individuals with psoriasis: a systematic review and meta-analysis. *J Investig Dermatol.* 2013;133(10):2340–6.
13. Liu L, Cui S, Liu M, Huo X, Zhang G, Wang N. Psoriasis increased the risk of adverse cardiovascular outcomes: a new systematic review and meta-analysis of cohort study. *Front Cardiovasc Med.* 2022;25(9):829709.
14. Khalid U, Ahlehoff O, Gislason GH, Kristensen SL, Skov L, Torp-Pedersen C, et al. Psoriasis and risk of heart failure: a nationwide cohort study. *Eur J Heart Fail.* 2014;16(7):743–8.
15. Mansouri B, Kivelevitch D, Natarajan B, Joshi AA, Ryan C, Benjegerdes K, et al. Comparison of coronary artery calcium scores between patients with psoriasis and type 2 diabetes. *JAMA Dermatol.* 2016;152(11):1244.
16. Lerman JB, Joshi AA, Chaturvedi A, Abera TM, Dey AK, Rodante JA, et al. Coronary plaque characterization in psoriasis reveals high-risk features that improve after treatment in a prospective observational study. *Circulation.* 2017;136(3):263–76.
17. Ahlehoff O, Gislason GH, Charlott M, Jørgensen CH, Lindhardsen J, Olesen JB, et al. Psoriasis is associated with clinically significant cardiovascular risk: a Danish nationwide cohort study: Psoriasis and cardiovascular risk. *J Intern Med.* 2011;270(2):147–57.
18. Mehta NN, Teague HL, Swindell WR, Baumer Y, Ward NL, Xing X, et al. IFN- γ and TNF- α synergism may provide a link between psoriasis and inflammatory atherogenesis. *Sci Rep.* 2017;7(1):13831.
19. Garshick MS, Barrett TJ, Wechter T, Azarchi S, Scher JU, Neimann A, et al. Inflammasome signaling and impaired vascular health in psoriasis. *Arterioscler Thromb Vasc Biol.* 2019;39(4):787–98.
20. Garshick MS, Tawil M, Barrett TJ, Salud-Gnilo CM, Eppler M, Lee A, et al. Activated platelets induce endothelial cell inflammatory response in psoriasis via COX-1. *Arterioscler Thromb Vasc Biol.* 2020;40(5):1340–51.
21. Von Stebut E, Reich K, Thaçi D, Koenig W, Pinter A, Körber A, et al. Impact of secukinumab on endothelial dysfunction and other cardiovascular disease parameters in psoriasis patients over 52 weeks. *J Investig Dermatol.* 2019;139(5):1054–62.
22. Teague HL, Varghese NJ, Tsoi LC, Dey AK, Garshick MS, Silverman JI, et al. Neutrophil subsets, platelets, and vascular disease in psoriasis. *JACC Basic Transl Sci.* 2019;4(1):1–14.
23. Vila L, Cullaré C, Solá J, Puig L, De Castellarnau C, De Moragas JM. Cyclooxygenase activity is increased in platelets from psoriatic patients. *J Investig Dermatol.* 1991;97(5):922–6.
24. Keermann M, Köks S, Reimann E, Abram K, Erm T, Silm H, et al. Expression of IL-36 family cytokines and IL-37 but not IL-38 is altered in psoriatic skin. *J Dermatol Sci.* 2015;80(2):150–2.
25. Berg AR, Hong CG, Svirydava M, Li H, Parel PM, Florida E, et al. Association of S100A8/A9 with lipid-rich necrotic core and treatment with biologic therapy in patients with psoriasis: results from an observational cohort study. *J Investig Dermatol.* 2022;142(11):2909–19.
26. Keermann M, Köks S, Reimann E, Prans E, Abram K, Kingo K. Transcriptional landscape of psoriasis identifies the involvement of IL36 and IL36RN. *BMC Genom.* 2015;16(1):322.
27. Köks S, Köks G. Activation of GPR15 and its involvement in the biological effects of smoking. *Exp Biol Med.* 2017;242(11):1207–12.
28. Krishnan VS, Köks S. Transcriptional landscape of repetitive elements in psoriatic skin from large cohort studies: relevance to psoriasis pathophysiology. *Int J Mol Sci.* 2023;24(23):16725.
29. Lättikivi F, Köks S, Keermann M, Reimann E, Prans E, Abram K, et al. Transcriptional landscape of human endogenous retroviruses (HERVs) and other repetitive elements in psoriatic skin. *Sci Rep.* 2018;8(1):4358.
30. Köks S, Keermann M, Reimann E, Prans E, Abram K, Silm H, et al. Psoriasis-specific RNA isoforms identified by RNA-Seq analysis of 173,446 transcripts. *Front Med [Internet].* 2016. <https://doi.org/10.3389/fmed.2016.00046/full>.
31. Kimball AB, Szapary P, Mrowietz U, Reich K, Langley RG, You Y, et al. Underdiagnosis and undertreatment of cardiovascular risk factors in patients with moderate to severe psoriasis. *J Am Acad Dermatol.* 2012;67(1):76–85.
32. Cea-Calvo L, Vanaclocha F, Belinchón I, Rincón Ó, Juliá B, Puig L. Underdiagnosis of cardiovascular

- risk factors in outpatients with psoriasis followed at hospital dermatology offices: the PSO-RISK study. *Acta Derm Venereol.* 2016;96(7):972–3.
33. Mehta NN, Li R, Krishnamoorthy P, Yu Y, Farver W, Rodrigues A, et al. Abnormal lipoprotein particles and cholesterol efflux capacity in patients with psoriasis. *Atherosclerosis.* 2012;224(1):218–21.
 34. Holzer M, Wolf P, Curcic S, Birner-Gruenberger R, Weger W, Inzinger M, et al. Psoriasis alters HDL composition and cholesterol efflux capacity. *J Lipid Res.* 2012;53(8):1618–24.
 35. Sorokin AV, Kotani K, Elnabawi YA, Dey AK, Sajja AP, Yamada S, et al. Association between oxidation-modified lipoproteins and coronary plaque in psoriasis: an observational cohort study. *Circ Res.* 2018;123(11):1244–54.
 36. Wu JJ, Poon KYT, Channual JC, Shen AYJ. Association between tumor necrosis factor inhibitor therapy and myocardial infarction risk in patients with psoriasis. *Arch Dermatol.* 2012;148(11):1244.
 37. Ports WC, Fayyad R, DeMicco DA, Laskey R, Wolk R. Effectiveness of lipid-lowering statin therapy in patients with and without psoriasis. *Clin Drug Investig.* 2017;37(8):775–85.
 38. Garshick MS, Baumer Y, Dey AK, Grattan R, Ng Q, Teague HL, et al. Characterization of PCSK9 in the blood and skin of psoriasis. *J Invest Dermatol.* 2021;141(2):308–15.
 39. Takeshita J, Wang S, Shin DB, Mehta NN, Kimmell SE, Margolis DJ, et al. Effect of psoriasis severity on hypertension control: a population-based study in the United Kingdom. *JAMA Dermatol.* 2015;151(2):161.
 40. Karbach S, Croxford AL, Oelze M, Schüler R, Minwegen D, Wegner J, et al. Interleukin 17 drives vascular inflammation, endothelial dysfunction, and arterial hypertension in psoriasis-like skin disease. *Arterioscler Thromb Vasc Biol.* 2014;34(12):2658–68.
 41. Langan SM, Seminara NM, Shin DB, Troxel AB, Kimmell SE, Mehta NN, et al. Prevalence of metabolic syndrome in patients with psoriasis: a population-based study in the United Kingdom. *J Invest Dermatol.* 2012;132(3):556–62.
 42. Elmetts CA, Leonardi CL, Davis DMR, Gelfand JM, Lichten J, Mehta NN, et al. Joint AAD-NPF guidelines of care for the management and treatment of psoriasis with awareness and attention to comorbidities. *J Am Acad Dermatol.* 2019;80(4):1073–113.
 43. Joshi AA, Lerman JB, Aberra TM, Afshar M, Teague HL, Rodante JA, et al. GlycA is a novel biomarker of inflammation and subclinical cardiovascular disease in psoriasis. *Circ Res.* 2016;119(11):1242–53.
 44. Naik HB, Natarajan B, Stansky E, Ahlman MA, Teague H, Salahuddin T, et al. Severity of psoriasis associates with aortic vascular inflammation detected by FDG PET/CT and neutrophil activation in a prospective observational study. *Arterioscler Thromb Vasc Biol.* 2015;35(12):2667–76.
 45. Reimann E, Lättekivi F, Keermann M, Abram K, Köks S, Kingo K, et al. Multicomponent biomarker approach improves the accuracy of diagnostic biomarkers for psoriasis vulgaris. *Acta Derm Venereol.* 2019;99(13):1258–65.
 46. Garshick MS, Ward NL, Krueger JG, Berger JS. Cardiovascular risk in patients with psoriasis. *J Am Coll Cardiol.* 2021;77(13):1670–80.
 47. Upala S, Sanguankee A. Effect of lifestyle weight loss intervention on disease severity in patients with psoriasis: a systematic review and meta-analysis. *Int J Obes.* 2015;39(8):1197–202.
 48. Ford AR, Siegel M, Bagel J, Cordoro KM, Garg A, Gottlieb A, et al. Dietary recommendations for adults with psoriasis or psoriatic arthritis from the medical board of the national psoriasis foundation: a systematic review. *JAMA Dermatol.* 2018;154(8):934.
 49. Dey AK, Joshi AA, Chaturvedi A, Lerman JB, Aberra TM, Rodante JA, et al. Association between skin and aortic vascular inflammation in patients with psoriasis: a case-cohort study using positron emission tomography/computed tomography. *JAMA Cardiol.* 2017;2(9):1013.
 50. Elnabawi YA, Dey AK, Goyal A, Groenendyk JW, Chung JH, Belur AD, et al. Coronary artery plaque characteristics and treatment with biologic therapy in severe psoriasis: results from a prospective observational study. *Cardiovasc Res.* 2019;115(4):721–8.
 51. Mehta NN, Shin DB, Joshi AA, Dey AK, Armstrong AW, Duffin KC, et al. Effect of 2 psoriasis treatments on vascular inflammation and novel inflammatory cardiovascular biomarkers: a randomized placebo-controlled trial. *Circ Cardiovasc Imaging.* 2018;11(6):e007394.
 52. Gelfand JM, Shin DB, Duffin KC, Armstrong AW, Blauvelt A, Tying SK, et al. A randomized placebo-controlled trial of Secukinumab on aortic vascular inflammation in moderate-to-severe plaque psoriasis (VIP-S). *J Invest Dermatol.* 2020;140(9):1784–1793.e2.

53. Penn Medicine [Internet]. 2024 [access 2025 July 3]. Program can identify and manage risk of cardiovascular disease in patients with psoriatic disease. <https://www.pennmedicine.org/news/program-for-psoriasis-patients-finds-cardiovascular-disease-risk>.
54. Armstrong EJ, Harskamp CT, Armstrong AW. Psoriasis and major adverse cardiovascular events: a systematic review and meta-analysis of observational studies. *J Am Heart Assoc*. 2013;2(2):e000062.
55. Mustonen A, Mattila K, Leino M, Koulu L, Tuominen R. How much of the productivity losses among psoriasis patients are due to psoriasis? *BMC Health Serv Res*. 2015;15(1):87.
56. Min S, Wang D, Xia J, Lin X, Jiang G. The economic burden and quality of life of patients with psoriasis treated with biologics in China. *J Dermatol Treat*. 2023;34(1):2247106.
57. Kruse CS, Argueta DA, Lopez L, Nair A. Patient and provider attitudes toward the use of patient portals for the management of chronic disease: a systematic review. *J Med Internet Res*. 2015;17(2):e40.
58. Stacey D, Lewis KB, Smith M, Carley M, Volk R, Douglas EE, et al. Decision aids for people facing health treatment or screening decisions. Cochrane Consumers and Communication Group, editor. *Cochrane Database Syst Rev* [Internet]. 2024. <https://doi.org/10.1002/14651858.CD001431.pub6>.
59. Smith A, Lai B, Zaman S, et al. Improving cardiovascular outcomes in the psoriasis cohort. Psoriasis and cardiovascular disease-clinician knowledge, practice and perceptions. *Australas J Dermatol*. 2025;66(4):e187–96. <https://doi.org/10.1111/ajd.14436>.
60. Afifi L, Shankle L, Armstrong AW, Boas M, Bridges A, Chiguil V, et al. National Psoriasis Foundation priorities for patient-centered research: proceedings from the 2016 conference. *J Psoriasis Psoriatic Arthritis*. 2017;2(3):73–80.

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